

Supplementary Materials

VariantFlow: a selective-execution engine for efficient population
genomic computation on large variant datasets

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This document is self-contained: each supplementary item (table or note) begins on its own page and restates the context needed to read it independently of the main text.

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Table S1: Cross-tool correctness validation

VariantFlow outputs were compared against VCFtools 0.1.17 and scikit-allel 1.3.13 on the IGSR 1000 Genomes chr22 high-coverage dataset (1,073,369 biallelic SNV records, 3,202 diploid samples). Correctness is assessed by exact output match (integer fields) or maximum absolute difference (floating-point fields). VariantFlow is judged correct only where the difference is zero (or within machine epsilon for floating-point).

Statistic	Comparison tool	Records	Max $ \Delta $	Exact?
Per-individual missingness (F_MISS)	VCFtools	3,202	0.0	Yes
Per-site missingness (F_MISS)	VCFtools	1,073,369	0.0	Yes
Allele frequency	VCFtools	925,730	0.0	Yes
Nucleotide diversity (π)	VCFtools	925,730	0.0	Yes
LD (r^2)	VCFtools	925,730	0.0	Yes
Weir-Cockerham F_{ST}	VCFtools	925,730	0.0	Yes
Per-individual missingness	scikit-allel	1,066,557	0.0	Yes
Site-frequency spectrum (folded and unfolded)	scikit-allel	5,000	0	Yes
π , d_{XY} pairwise counts (15% missing)	pixy	5,000	0	Yes

VCFtools rows are byte-identical after normalisation (Note S2). The scikit-allel missingness and site-frequency-spectrum rows are exact: maximum absolute difference 0. For pixy, the integer pairwise counts on which the estimator is defined are identical in every window, and the derived window averages agree to within 5×10^{-9} . Full cross-check outputs are recorded in `benchmark/reports/v30-correctness-crosschecks.md`. No scikit-allel cross-check of allele frequency was performed; allele-frequency correctness rests on the VCFtools parity check.

Table S2: Benchmark system configuration

Benchmarks were measured on more than one host, and host choice is material: VCFtools linkage disequilibrium takes 796 s on the Docker/Linux host and 445 s on the Apple M5 Max, a 1.8-fold difference. The population-genetic results in Table S3 were measured on the M5 Max configuration below. The index-assisted filtering and Parquet/DuckDB results were measured on a Docker/Linux host, and the CHM13 FORMAT-filter benchmark used bcftools 1.23.1 rather than the version listed below. Host, tool versions, and the exact commands are recorded per row in the corresponding benchmark report. Timing is wall-clock, single-threaded; peak memory is peak resident set size (RSS).

Parameter	Value
Hardware	Apple M5 Max, 14-core CPU, 128 GB unified memory
OS	macOS 15.x (arm64)
VariantFlow	v1.5.0 (Zenodo doi:10.5281/zenodo.21198172)
bcftools	1.21 (htslib 1.21) for the indexed-filter benchmarks; 1.23.1 for the CHM13 FORMAT-filter benchmark
VCFtools	0.1.17
scikit-allel	1.3.13
pixy	2.2.1
Input data	IGSR 1000 Genomes high-coverage (GRCh38); DDBJ CHM13 chr22 (T2T)
chr22 (IGSR)	1,073,369 biallelic records, 3,202 samples
chr1 (IGSR)	6,468,094 biallelic records, 3,202 samples
CHM13 chr22 (DDBJ)	3,715 samples; T2T-CHM13 joint call (FORMAT-filter benchmark)
pixy validation set	5,000 all-sites records, 40 samples, 2 populations, 15% missing
Memory metric	Peak RSS via <code>resource.getrusage(RUSAGE_CHILDREN)</code>
Timing	Wall-clock seconds, single-threaded execution

Table S3: Timing and memory for streaming and repeated-query workflows

Per-workflow wall-clock time and peak memory for VariantFlow and the baseline tool, on the system described in Table S2. Baseline is VCFtools 0.1.17 for all population-genetics rows. Each workflow was correctness-checked against its reference before timing (Table S1). Timings for the FORMAT-filter and indexed predicate workflows are summarised in Table 1 of the main text.

Workflow	Dataset	VariantFlow (s)	VCFtools (s)	VariantFlow (MB)	VCFtools (MB)
Missingness	chr22 full (1.07M)	23.75	201.67	8.6	4.1
Missingness	chr1 full (6.47M)	128.27	1031.13	8.7	4.1
LD (r^2)	chr22 (925k)	75.9	444.9	9.0	4.2
Frequency	chr22 (925k)	23.56	86.77	8.7	4.1
Heterozygosity	chr22 (925k)	96.09	140.16	9.3	4.3
Hardy-Weinberg	chr22 (925k)	85.77	105.55	9.0	4.7
π (per-site)	chr22 (925k)	86.11	92.70	9.0	4.1
Tajima's D	chr22 (925k)	100.15	127.05	27.1	13.8
F_{ST} (Weir-Cockerham)	chr22 (925k)	84.90	92.14	9.3	4.7

Measured on the Apple M5 Max (Table S2), VariantFlow 1.5.0 vs VCFtools 0.1.17, single-threaded wall-clock; RSS = peak resident set size. Replicates: each chromosome 22 row is the faster of two runs, LD is the median of five isolated runs (spread 74.0–79.8 s), and the chromosome 1 row is a single run. Speedups quoted in the main text are baseline/VariantFlow ratios. VariantFlow computes site and individual missingness in one pass, so the missingness baseline sums VCFtools `-missing-site` and `-missing-indv`. F_{ST} uses a deterministic 50/50 sample split (identical for both tools), and LD is bounded at `-max-distance 500` for both tools. “925k” is the 925,730-record biallelic-SNV chromosome 22 subset; missingness is timed on the full biallelic chromosome. Every row was output-parity checked against VCFtools (Table S1).

Repeated-query workflow (Parquet export + DuckDB). The export-once, query-many workflow was benchmarked separately because its cost model differs from a single streaming pass: the VCF is converted to Parquet once, after which repeated analytical queries run against the columnar file. The measurements below use a deterministic stress VCF (DuckDB 1.5.2, bcftools 1.21, Parquet row-group size 65 536, five repeated queries; each time is the wall-clock mean of three runs). The *query-only* speedup is the steady-state ratio once the Parquet file exists; the *amortized* speedup charges the full one-time export against a single query, a conservative figure that approaches the query-only ratio as queries are repeated. Both ratios are reproducible from the three time columns. Every query returned counts identical to the bcftools baseline.

Query	Records	Export (s)	DuckDB (s)	bcftools (s)	Query-only	Amortized
QUAL>30	100 000	0.091	0.068	0.935	13.74x	5.87x
QUAL>30	1 000 000	0.767	0.068	8.444	124.41x	10.12x
INFO/DP>40	100 000	0.085	0.061	1.605	26.27x	10.97x
INFO/DP>40	1 000 000	0.749	0.068	15.691	231.60x	19.20x
FILTER=PASS	100 000	0.085	0.066	0.899	13.58x	5.94x
FILTER=PASS	1 000 000	0.763	0.070	9.307	133.66x	11.17x
group-by CHROM	100 000	0.092	0.082	0.207	2.51x	1.19x
group-by CHROM	1 000 000	0.766	0.131	1.818	13.92x	2.03x

`bcftools` (s) is the repeated-scan baseline for the same query workload. Query-only speedup = `bcftools` / DuckDB; amortized speedup = `bcftools` / (export + DuckDB). Amortized speedup ranges from 1.19x (grouped **CHROM** counts, 100k records) to 19.20x (**INFO/DP>40**, 1M records), the 1.19–19.20x range quoted in the main text. Grouped counts amortize weakly because a single grouped scan in `bcftools` is already cheap; the workflow’s advantage grows with query selectivity and repetition.

Table S4: Missing-data validation against pixy

The `pixy` tool (Korunes and Samuk, 2021) provides unbiased estimators for nucleotide diversity (π) and between-population divergence (d_{XY}) that account for missing genotypes by counting only non-missing pairwise comparisons at each site, aggregated as a ratio of sums across all sites in a window (including invariant sites). VariantFlow implements the identical estimator in its `pixy` subcommand (Note S1). To validate this, we generated a deterministic all-sites VCF (5,000 sites, 40 diploid samples, two populations) with 15% of genotypes set to missing, and computed windowed π and d_{XY} (1 kb windows) with both tools.

Statistic	Comparison	Windows	Count match	Max $ \Delta $
π (per population)	<code>pixy</code> 2.2.1	10	Exact	$< 5 \times 10^{-9}$
d_{XY} (per pop. pair)	<code>pixy</code> 2.2.1	5	Exact	$< 5 \times 10^{-9}$

Integer pairwise counts (`count_diffs`, `count_comparisons`) are byte-identical to `pixy` across every window; the residual $|\Delta|$ in the averaged ratio reflects only the eight-decimal print precision of VariantFlow. The comparison is performed on data containing missing genotypes, the regime `pixy` was designed for.

Table S5: Statistics coverage relative to scikit-allel

Population-genetics statistics grouped by computational class. The organizing rule is the memory-access pattern each statistic requires. **Group A** statistics are computable in a single streaming pass over sites or windows and are implemented by VariantFlow’s selective-execution engine; overlapping statistics are benchmarked against scikit-allel (Miles et al., 2024) and/or VCFtools. **Group B** statistics require the full genotype or phased haplotype matrix to be held in memory simultaneously; these are the domain of matrix-oriented libraries such as scikit-allel, and VariantFlow defers to them by design rather than reimplementing them. This delineation reflects VariantFlow’s scope: it complements, and does not replace, scikit-allel.

Statistic	VariantFlow	scikit-allel name	Validated
<i>Group A — site-wise / window-wise (streaming; VariantFlow’s domain)</i>			
Allele frequency / counts	freq	allele_frequencies	✓ (VCFtools, s-a)
Genotype missingness	missingness	(genotype masks)	✓ (VCFtools, s-a)
Observed/expected heterozygosity	het	heterozygosity_*	VCFtools
Hardy–Weinberg equilibrium	hardy	—	VCFtools
Nucleotide diversity π	pi	sequence_diversity	✓ (VCFtools)
Missing-data-aware π , d_{XY}	pixy	(pixy)	✓ (pixy, exact)
Tajima’s D	tajima-d	tajima_d	VCFtools
Watterson’s θ	tajima-d [†]	watterson_theta	partial
Site frequency spectrum (folded/unfolded)	sfs	sfs, sfs_folded	✓ (s-a, exact)
F_{ST} (Weir–Cockerham, Hudson)	fst	*_fst	✓ (VCFtools)
Linkage disequilibrium r^2	ld	rogers_huff_r	✓ (VCFtools)
<i>Group B — matrix-wide / haplotype-based (defer to scikit-allel by design)</i>			
Joint / scaled SFS (2D)	—	joint_sfs, sfs_scaled	—
Selection scans (EHH, iHS, XP-EHH, nSL)	—	ehh, ihs, ...	—
Garud’s H , haplotype diversity	—	garud_h, ...	—
D -/ f -statistics (ABBA-BABA, $f_{2/3/4}$)	—	patterson_d, ...	—
Distance, PCoA, PCA	—	pca, pcoa, ...	—

s-a = scikit-allel. ✓ = output validated (exact or within machine epsilon) against the named tool. A bare tool name (no ✓) means the output was compared against that tool and agreed but is not among the quantified rows of Table S1; “partial” means only the Watterson’s θ component of the `tajima-d` output is exercised. [†]Watterson’s θ is reported as a component of the `tajima-d` windowed output. Group B statistics require the whole genotype/haplotype matrix in memory and are outside VariantFlow’s streaming scope; users should use scikit-allel for these.

Note S1: Missing-data-aware π and d_{XY}

VariantFlow's `pixy` subcommand consumes an all-sites VCF and, at each site, tallies pairwise allelic differences and comparisons over non-missing gametes only: for a population with per-allele counts c_a and $n = \sum_a c_a$ non-missing gametes, comparisons = $\binom{n}{2}$ and differences = $(n^2 - \sum_a c_a^2)/2$; for a population pair with counts $c_a^{(1)}, c_a^{(2)}$, comparisons = $n_1 n_2$ and differences = $n_1 n_2 - \sum_a c_a^{(1)} c_a^{(2)}$. Windowed π and d_{XY} are the ratio of summed differences to summed comparisons over all sites in the window. Because invariant sites contribute comparisons with zero differences and missing genotypes are excluded per site, the estimator is unbiased under non-uniform missingness, matching `pixy` (Korunes and Samuk, 2021) exactly (Table S4) while running as a compiled selective-execution pass.

Note S2: Correctness validation methodology

Each benchmark row in Table 1 of the main text undergoes correctness validation before any speedup claim is reported. The validation process:

1. Run VariantFlow and the baseline tool (bcftools or VCFtools) on identical input with equivalent parameters.
2. Normalize outputs: strip headers, sort by genomic position, unify column separators.
3. Compute `diff` on normalized outputs. A row earns ✓ only if the diff is empty (zero differing lines).
4. For floating-point statistics (e.g., F_{MISS} , π), additionally verify maximum absolute difference $< 10^{-10}$.

For the scikit-allel cross-check, we independently compute per-individual missingness, allele frequency, and the site frequency spectrum from the genotype matrix and compare against VariantFlow’s output. This provides a third-party validation independent of both VariantFlow and VCFtools codebases.

References

- Korunes, K. L. and Samuk, K. (2021). pixy: Unbiased estimation of nucleotide diversity and divergence in the presence of missing data. *Molecular Ecology Resources*, 21:1359–1368.
- Miles, A. et al. (2024). scikit-allel: a Python package for exploring and analysing genetic variation data. <https://github.com/cggh/scikit-allel>.